

Efficient coding:

Horace Barlow proposed the efficient coding hypothesis in 1961. It's been hugely influential because it posits why visual systems work as they do.

In Barlow's words, "A wing would be a most mystifying structure if one did not know that birds flew... What would be a similar summary of the most important operation performed at a sensory relay?"

He put forward 3 hypotheses.

1. Detecting "passwords" of key significance
 2. Filtering signals for other parts of nervous system.
 3. Recoding sensory messages from redundant inputs.
- Hypothesis 3 is efficient coding. The basic idea is that sensory systems shouldn't waste energy and spikes to transmit redundant information.

A precise formulation requires information theory.

Information theory basics

Consider a discrete random variable

$$x \in X = \{x_1, x_2, \dots, x_n\}$$

$$x \sim P(x)$$

We first want to quantify how surprising it is to observe $x = x_i$. It is a function of the probability.

$$1. P(x_i) = 1 \Rightarrow \Delta(p_i) = 0$$

$$2. P(x_i) < P(x_j) \Rightarrow \Delta(p_i) > \Delta(p_j)$$

$$3. P(\{x_i, x_j\}) = P(x_i)P(x_j) \Rightarrow \Delta(p_i p_j) = \Delta(p_i) + \Delta(p_j)$$

The function satisfying this is

$$\Delta(p) = \log\left(\frac{1}{p}\right) = -\log p$$

Where the base of the logarithm is arbitrary: $\log_2 \Rightarrow$ bits, $\ln \Rightarrow$ nats, etc. The expected surprise is called the entropy.

$$H(x) = -\sum_{x \in X} P(x) \log P(x)$$

It quantifies x 's level of uncertainty.

Now consider a second random variable

$$y \in Y = \{y_1, y_2, \dots, y_m\}$$

$$x, y \sim P(x, y)$$

The expected surprise of observing y given x is called the conditional entropy

$$H(y|x) = -\sum_{x \in X} \sum_{y \in Y} P(x, y) \log P(y|x)$$

The mutual information between X and Y is how much your expected surprise about one decreases by knowing the other.

$$I(x; y) = H(y) - H(y|x)$$

$$= \sum_{x \in X} \sum_{y \in Y} P(x, y) \log \frac{P(y|x)}{P(y)}$$

$$= \sum_{x \in X} \sum_{y \in Y} P(x, y) \log \frac{P(x, y)}{P(x)P(y)} = H(x) + H(y) - H(x, y)$$

$$= \sum_{x \in X} \sum_{y \in Y} P(x, y) \log \frac{P(x|y)}{P(x)} = H(x) - H(x|y)$$

The mutual information provides a general notion of statistical dependence

$$I(x; y) \geq 0, \quad I(x; y) = 0 \iff P(x, y) = P(x)P(y)$$

These notions can be generalized to continuous random variables

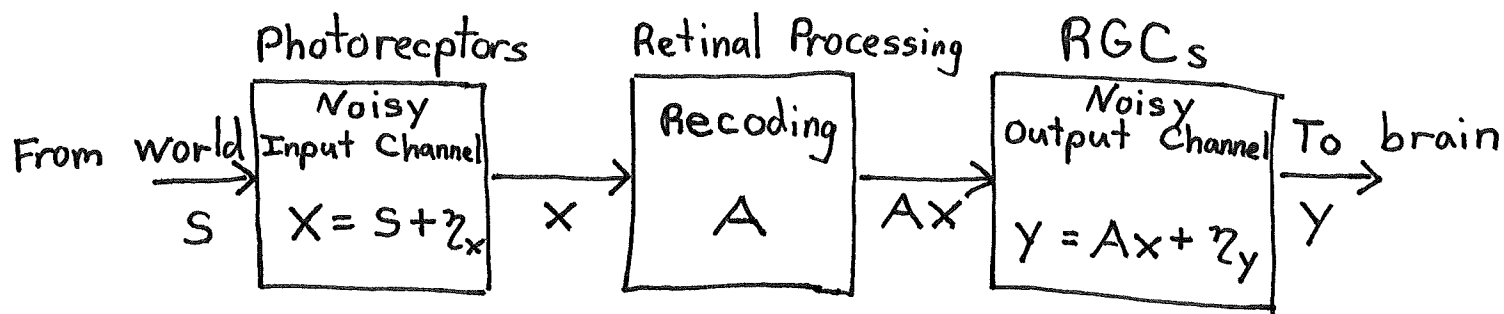
$$x \in [a, b], \quad x \sim \varrho(x)$$

$$\mathcal{H}(x) = - \int_a^b dx \varrho(x) \log \varrho(x)$$

This produces a nice notion of mutual information for continuous variables, but be careful interpreting $\mathcal{H}$ directly. For example, it can be negative, conflicting with the notion of expected surprise.

Efficient coding in retina:

In 1990, Atick and Redlich proposed a beautiful efficient coding theory for RGCs.



As schematized above, they conceptualized the retina as a noisy processor that detects sensory signals, s , and recodes them for further processing. They modeled the signal and noise as Gaussian.

$$s \sim \mathcal{N}(0, C_s), \quad z_x \sim \mathcal{N}(0, \sigma_x^2), \quad z_y \sim \mathcal{N}(0, \sigma_y^2)$$

The signal differs from noise because it's correlated

$$(C_s)_{ij} = \langle s_i s_j \rangle = \underbrace{s^2}_{\text{Signal magnitude}} \underbrace{e^{-\|\underline{p}_i - \underline{p}_j\| / \underline{D}}}_{\text{Exponential falloff with distance between points}} \underbrace{1}_{\text{Length scale}}$$

From these quantities, one sees

$$X \sim \mathcal{N}(0, C_x), \quad Y \sim \mathcal{N}(0, C_y)$$

$$C_x = C_s + \sigma_x^2 \mathbf{I}$$

$$(C_y)_{ij} = \langle Y_i Y_j \rangle = \langle (Ax)_i (Ax)_j \rangle + \sigma_y^2 \delta_{ij}$$

$$= \left\langle \sum_k \sum_\ell A_{ik} X_k A_{j\ell} X_\ell \right\rangle + \sigma_y^2 \delta_{ij}$$

$$= \sum_k \sum_\ell A_{ik} \underbrace{\langle X_k X_\ell \rangle}_{C_x} A_{j\ell} + \sigma_y^2 \delta_{ij}$$

$$C_y = A C_x A^T + \sigma_y^2 \mathbf{I}$$

$$C_y = A(C_s + \sigma_x^2 I)A^T + \sigma_y^2 I$$

With these Gaussian assumptions, the mutual information between the RGC output and the signal is

$$I(y; s) = \frac{1}{2} \log \left[\frac{\det(A(C_s + \sigma_x^2 I)A^T + \sigma_y^2 I)}{\det(A\sigma_x^2 I A^T + \sigma_y^2 I)} \right]$$

Atick and Redlich then made two hypotheses. First, the ~~original~~ ^{recoding, A,} must retain a certain amount of information, I^* , "to meet an organism's needs."

$$\underbrace{I(y; s) = I^*}_{\text{Constraint equation on A}} \leq \underbrace{I(x; s)}_{\text{Information processing inequality}}$$

Second, the recoding, A, should minimize the redundancy

$$R = 1 - I(y; s) / C_{\text{out}}(y)$$

$$\underbrace{C_{\text{out}}(y)}_{\text{channel capacity of output}} = \max_{P(w)} I(y; w) \Big|_{\substack{\langle y^2 \rangle = \text{const.} \\ \text{Fixed RGC output level}}}; \quad y = w + \eta_y$$

The channel capacity can be evaluated by noting

$$I(y; w) = \frac{1}{2} \log \left[\frac{\det(C_w + \sigma_y^2 I)}{\det(\sigma_y^2 I)} \right]$$

is maximized when C_w , and thus C_y , are diagonal (subject to the $\langle y^2 \rangle = \text{const.}$ constraint). In other words, the redundancy objective says that $I(y; s)$

should be as close to possible to what you'd get with decorrelated RGCs. However, maybe decorrelation also decreases information, so you also have to ensure that you maintain I^* .

Solution: Through more complicated math and numerical optimization, Atick and Redlich solved this optimization problem

$$\hat{A} = \arg \min_A (R \mid I(y; s) = I^*)$$

and found center-surround receptive fields. The solution wasn't unique though,

$$\underbrace{U^T U}_{\text{orthogonal matrix}} = I \Rightarrow U \hat{A} \text{ is also optimal}$$

Center-surround receptive fields were the only rotationally symmetric solution.